Project Documentation: Military Analytics & Global Power Indexing

Internship Program: Infosys Springboard

Milestone: 2 (Data Processing & KPI Generation)

Project Name: Unified Military Analytics and Visualization

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1. Project Overview

This project aims to analyze global military strength by integrating diverse datasets including manpower, equipment, logistics, and economic factors. Milestone 2 focuses on **Feature Engineering** and **KPI (Key Performance Indicator) Calculation** to transform raw data into a normalized "Overall Power Index."

2. Data Processing & Methodology

The data processing was performed using Python (Pandas) in a Google Colab environment. The core methodology involves categorizing raw military metrics and applying **Min-Max Normalization** to ensure all features are comparable on a scale of 0 to 1.

A. Power Categorization

The following pillars were established to define a nation's strength:

- **Manpower:** Active/reserve personnel and population fit for service.
- **Air Power:** Total aircraft, fighters, and attack helicopters.
- **Land Power:** Tanks, armored vehicles, and artillery.
- **Naval Power:** Fleet strength, tonnage, and specialized vessels (Carriers/Submarines).
- **Logistics:** Infrastructure including ports, airports, and transport networks.
- **Energy:** Oil, coal, and gas production and reserves.
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B. KPI Engineering (Code Logic)

Specific formulas were implemented to derive advanced insights:

1. **Debt Management:** Calculated as the ratio of external debt to Purchasing Power Parity (PPP).
2. $Debt\ Ratio = (External\ Debt\ (USD)) / (Purchasing\ Power\ Parity\ (USD))$

3. **Budget Efficiency:** Defense budget relative to the country's PPP(Purchasing Power Parity).
4. **Assets Per Capita:** A normalization of total military assets divided by the total population to measure military density.
5. **Alliance Tagging:** Automated classification of countries into **NATO**, **EU**, and **BRICS** for geopolitical comparison.

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3. Feature Selection & Final Indexing

After generating raw sums and normalized scores, redundant raw columns were dropped to optimize the dataset for visualization. The **Overall Power Index** was calculated as the mean of all normalized power pillars:

$$Overall\ Power\ Index = \frac{\sum_{i=1}^{10} \text{Normalized Pillar}_i}{10}$$

A final **Overall Rank** was assigned using a dense ranking method, where a lower rank (e.g., 1) indicates a higher power index.

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4. Power BI Visualization (Milestone 2 Integration)

The processed new_kpi_df was exported to Power BI to create an interactive "Nation Overview" dashboard.

Feature	**Visualization Type**	**Purpose**
Quick Stats	Cards / Gauges	Instant view of a nation's Rank and Power Index.
Power Distribution	Pie Charts	Breaks down how much a specific category (e.g., Economic Power) contributes to the total.
Comparative Strength	Bar Charts	Ranks countries based on Assets Per Capita or Naval Power.
Geopolitical Analysis	Slicers	Filter data by NATO, EU, or BRICS status.

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5. Conclusion & Next Steps

The completion of Milestone 2 provides a structured, cleaned, and ranked dataset ready for deep analytical exploration.

- **Current Achievement:** Successfully transformed 50+ raw metrics into a unified 0-1 Power Index.